In silico first identification and modelling of DBP hydrolases in the marine diatom *Cylindrotheca closterium*

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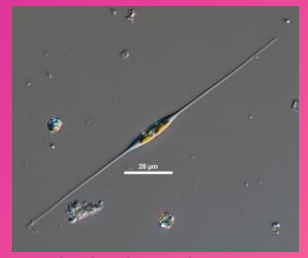
1. INTRODUCTION

Phthalate esters (PEs) are the most common plasticizers worldwide, even if some of them are classified as endocrine disrupting chemicals (EDCs) and their presence in the human body has been linked to different diseases. Enzymes involved in the degradation of PEs have been found in different bacteria but never in microalgae.

The marine diatom ***Cylindrotheca closterium*** was found effective in di-n-butyl-phthalate (DBP) degradation in sediments *[Zhang, 2019]*, and its transcriptome, produced and annotated by our group *[Elagoz, 2020]* was analyzed in order to search of transcripts related to such activity.



Cylindrotheca closterium

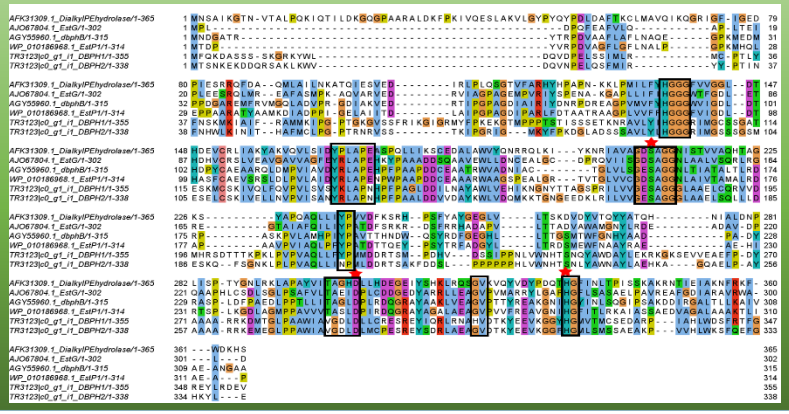
2. TRANSCRIPTOME MINING

Tranciptome mining with **Class IV PE-Esteases** → **"TR3123|c0_g1_i1 len=2609"** → Two CDS (***dbph1* and *dbph2***)

DBPH1 putative catalytic triad: S180, H323, D293
DBPH2 putative catalytic triad S170, H309, D279

ENZYME	HGGG	GXSAGG	HG	TH	YP	GV	YXLAPE	TAX XD
DBPH1	HGGG	GESAGG	HG	TE	YP	HV	YKLAPN	VGD LD
DBPH2	HGGG	GESAGG	HG	ME	NP	GV	YRLAPN	VGD LD

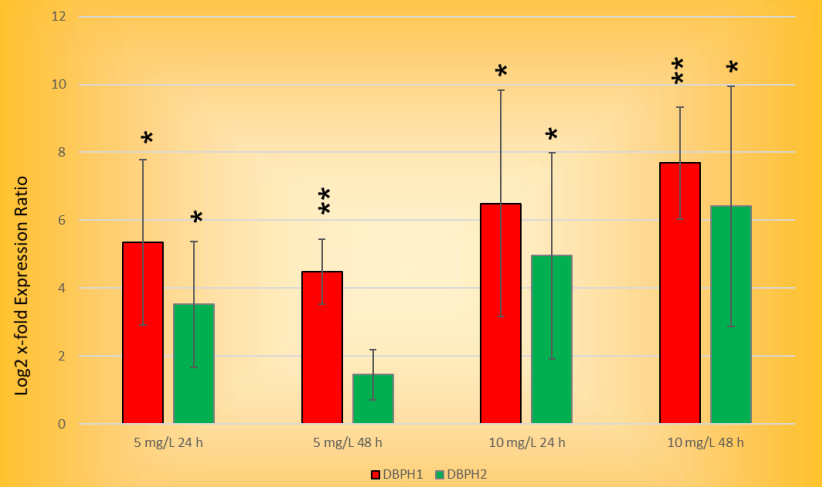
Acid Residue



!FUTURE APPROACHES!

- Diatom esterases conserved motifs identification
- Screening and RT-qPCR of other PE-hydrolases in *C. closterium*
- First PE-hydrolase 3D structure resolution

3. *dbph1* and *dbph2* RELATIVE EXPRESSION ANALYSIS

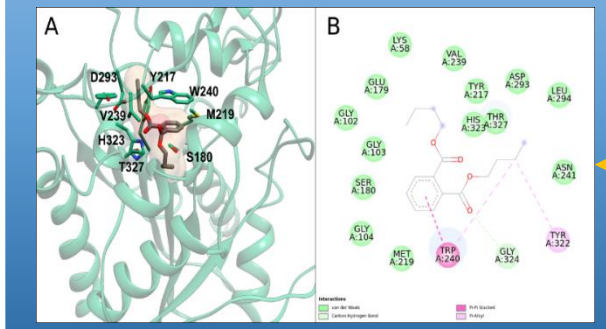


- *dbph1* and *dbph2* expression levels are **time and concentration-dependent**
- *dbph1* is **more highly expressed** than *dbph2*

** = p<0.05; ** = p<0.01*

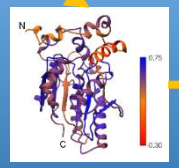
4. MODELLING & DBPH1 DOCKING WITH DBP

DBPH1-DBP binding via AutoDock vina:
- Favorable affinity **(-31.0 kJ/mol)**
- Binding site with the catalytic triad
- Significant energetic contribution provided by **W240** (pi-interaction, as in *[Sarkar et al.2020]*)



I-TASSER iterative fragment assembly approach

DBPH1
c-score: 0.22
Removal of residues 1-20



DBPH2
c-score: 0.38
Removal of residues 1-40

DBPH2-DBP binding via AutoDock vina: